Network Science

Lecture 08: Network Dynamics

5942UE Network Science

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Network Dynamics

This lecture moves from **static structure** to **processes that unfold on networks over time**.

Focus

- diffusion as a network process
- epidemic and contagion intuition
- threshold cascades and coordination
- temporal networks and changing structure

Module 1

Diffusion

Guiding Question

How does activity move through a network once a process has started?

$$\text{diffusion} = \text{state change propagated along edges}$$

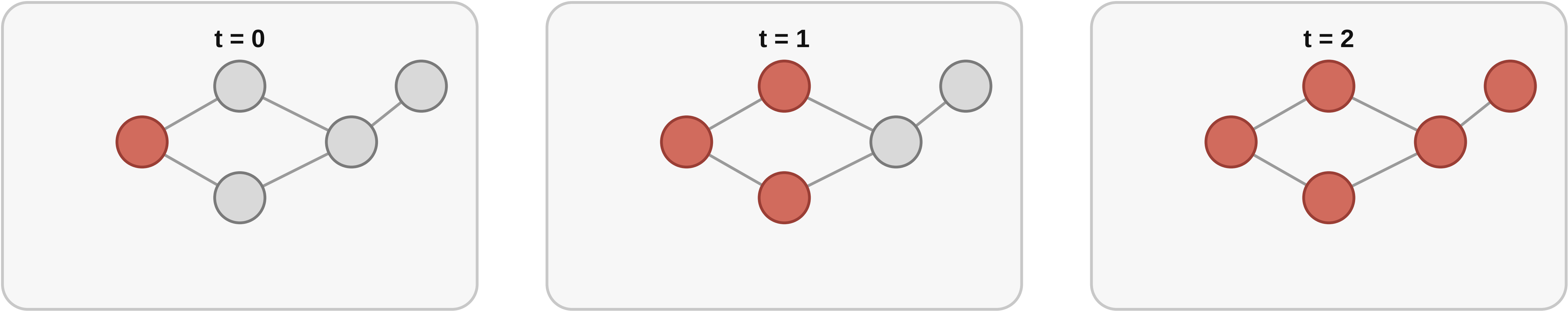
Formal Intuition

In a diffusion process, nodes have **states** and edges define **who can influence whom**.

At each step, active nodes may activate their neighbors according to a rule.

Example Process

Diffusion over time



Red nodes are active; each step spreads activity to reachable neighbors.

Interpretation

- paths constrain where influence can go
- degree and centrality affect how quickly activity spreads
- bottlenecks and components can block diffusion entirely

> Diffusion is not just about *who* is active. It is about how the network *channels* activation over time.

Module 2

Epidemics and Simple Contagion

Guiding Question

What changes when contagion spreads through repeated contact on a network?

Formal Intuition

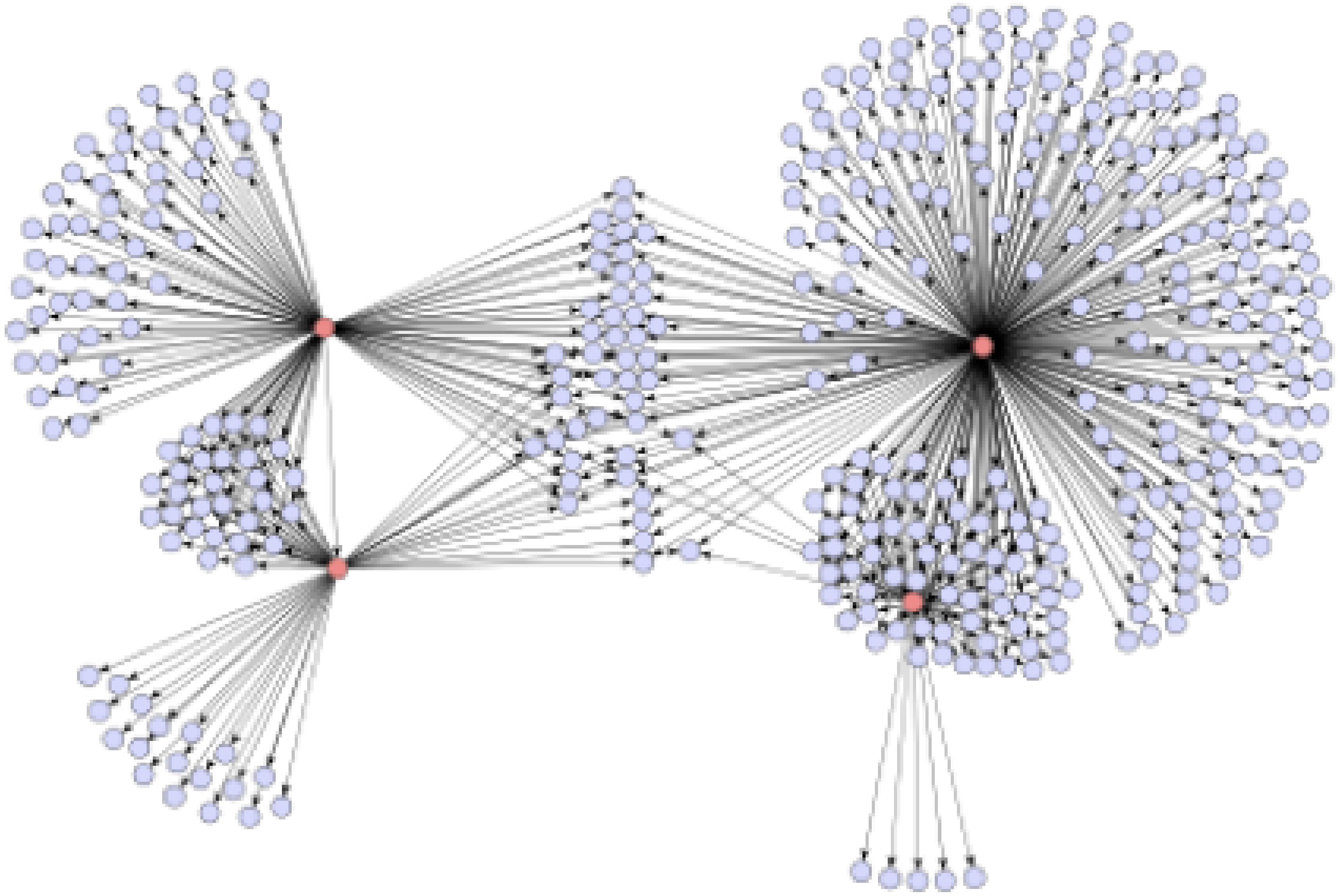
Simple contagion treats activation as transmission through individual contacts.

Examples include:

- infection
- message forwarding
- some forms of information exposure

Epidemic Motivation

Cascading Communication



Interpretation

The same broad network idea can describe both **disease spread** and **information forwarding**, but the process rules are not identical.

> Networks make contagion structured: spread depends on contact pattern, repetition, and the local transmission rule.

Module 3

Cascades and Thresholds

Guiding Question

Why do some behaviors need reinforcement from several neighbors before they spread?

$$\text{adopt if enough neighbors adopted}$$

Threshold Rule

Each node adopts only when the **share of active neighbors** reaches a threshold.

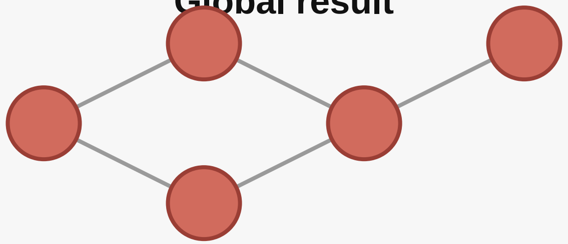
Cascade Example

Threshold cascade

Local rule

```
activate v if
active neighbors of v
---------------------- >= theta
total neighbors of v
```

Global result



A local adoption threshold can trigger a global cascade.

Interpretation

- this is more than simple one-contact transmission
- **dense local reinforcement** can help adoption start
- bridges matter differently than in simple contagion

> Cascades depend not only on exposure, but on how much local confirmation a node requires before changing state.

Module 4

Temporal Networks

Guiding Question

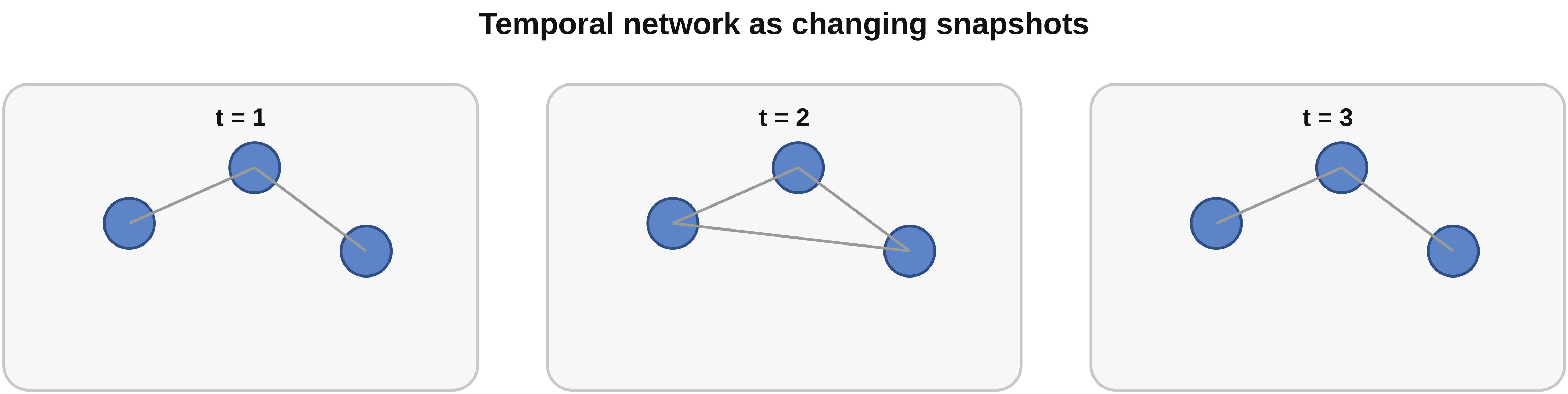
What do we miss when we replace an evolving network with one static snapshot?

Formal Intuition

A **temporal network** records *when* edges exist, not just *whether* they exist.

That means reachability can depend on **order** as well as connectivity.

Snapshot View



A static graph is one slice; temporal networks track when edges appear, disappear, or reorder.

Interpretation

- the same nodes can have different paths at different times
- diffusion may succeed or fail depending on timing
- static aggregation can hide causal order

> Temporal networks extend graph theory by making change and order explicit rather than treating them as noise.

Wrap-Up

- dynamics depend on both network structure and process rules
- simple contagion and complex contagion behave differently
- timing matters: the same graph can support different outcomes when edges change